

PHY102 Electricity and Magnetism Jan-April 2026: Assignment 9

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To be discussed in the Tutorial on April 17, 2026

1. Consider a system of two concentric current rings of radii r_1 and r_2 laying in the same plane. Calculate the mutual inductance of this system.
2. Consider a coil of radius r with n turns. What is the self inductance of this coil?
3. Consider a long solenoid (infinite) with n turns per unit length and current I flowing through it. What is the magnetic field inside the solenoid? Imagine that you now go to a frame of reference moving with velocity v in the z direction. By transforming the source to the new frame, calculate the magnetic field seen in the moving frame.
4. Consider a region of space with no magnetic field, however having some electric field. What will be the magnetic field in this region seen from other frames of reference? How exactly the electric and magnetic fields will be related to each other in arbitrary frames? Repeat the calculation with a region with no electric field but non-zero magnetic field in some frame. What will the relationship be between the electric and magnetic field in arbitrary frames for this case?